



Contents lists available at ScienceDirect

Engineering

journal homepage: www.elsevier.com/locate/eng

Research
Intelligent Manufacturing—Article

IoT-Enabled Real-Time Energy Consumption Anomaly Detection and Diagnosis for Automotive Paint Drying System

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ARTICLE INFO

Article history:

Received 30 July 2024

Revised 21 August 2025

Accepted 30 September 2025

Available online xxx

Keywords:

Intelligent sustainable manufacturing
Energy consumption anomaly detection
Anomaly diagnosis
Automotive paint drying system
Deep Learning

ABSTRACT

The energy-intensive automotive industry requires sophisticated energy management systems to improve energy efficiency. In automotive workshops, paint drying systems are a significant energy consumer, necessitating real-time monitoring and control to minimize energy waste and potentially prevent system malfunctions. Thus, this study proposed a novel real-time energy consumption anomaly detection and diagnosis methodology (eAnoD) for automotive paint drying systems to enhance their energy efficiency and operational safety. Specifically, an architecture combining a temporal convolutional network and graph attention network (TCN-GAT) was devised to extract spatiotemporal features from multidomain data, including energy consumption, equipment parameters, production states, and environmental conditions. A hybrid neural network combining a backpropagation neural network (BPNN) and variational autoencoder (VAE) was constructed to enable the prompt identification of energy consumption deviations. Furthermore, an anomaly grading method integrating combination weighting and cloud modeling techniques was developed to evaluate anomaly severity, facilitating targeted maintenance and proactive risk prevention. A real-world case study was conducted in a new-energy vehicle factory to validate the effectiveness and practicality of the proposed methodology and demonstrate its potential for energy saving and risk mitigation in automotive manufacturing. This study is expected to serve as a reference for practical implementation and generate new ideas for academic exploration.

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1. Introduction

Over the past three decades, rapid global economic growth has doubled energy consumption, with the manufacturing sector responsible for 54% of this increase [1]. As a significant energy consumer in this sector, the automotive industry faces escalating pressure to improve energy efficiency, a goal further underscored by Industry 5.0, a paradigm that emphasizes industrial sustainability [2]. Driven by this imperative, the automotive manufacturing industry is actively pursuing digital, intelligent, and green transformations [3,4]. Although the adoption of smart sensors and Internet of Things (IoT) technologies has enabled real-time energy data monitoring in some facilities [5], the vast potential of these data

for energy optimization remains largely untapped because many automotive manufacturers lack a clear roadmap for achieving effective energy savings. Consequently, this emphasizes the critical need for further studies to explore the value of real-time data in delivering tangible energy efficiency improvements.

Among the four primary automotive manufacturing workshops, paint shops are the most energy intensive, consuming up to 70% of the total energy used [6]. The drying system, which operates under stringent high-temperature and high-pressure conditions, is the largest energy consumer in paint shops. These demanding operating conditions increase the likelihood of energy consumption anomalies, which, if undetected and unaddressed, can compromise paint quality, reduce energy efficiency, increase production costs, and pose safety hazards. For instance, deviations from the expected energy consumption patterns during reheating (e.g., unusual increases) or cooldown (e.g., unexpectedly slow decreases) can indicate underlying issues such as faulted heating elements,

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<https://doi.org/10.1016/j.eng.2025.09.034>

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cooling system inefficiencies, or insulation degradation. In a drying system, a seemingly minor deviation in energy consumption, if left unaddressed, can escalate into significant problems that affect both efficiency and safety. Under these circumstances, real-time energy anomalies and diagnoses can serve as early indicators of potential system operation faults and enable predictive maintenance strategies.

However, although individual devices log most anomaly records, these logs are inaccessible through a centralized interface and must be manually downloaded from each device. This constraint prevents the real-time anomaly detection. Moreover, automated and programmatic access is hampered by restricted application programming interface (API) access to proprietary systems that are strictly managed by suppliers. Although event logs capture a wide range of error types, not all are related to anomalies from the perspectives of energy efficiency and safe operation. Therefore, the ability to detect and diagnose potential system malfunctions in real-time using multidomain data is crucial to ensure effective predictive maintenance, which ultimately optimize energy efficiency, maximize production uptime, and ensure a safe operating environment.

Although approaches for energy consumption anomaly detection and diagnosis have been explored in sectors such as buildings and power systems [7,8], research that specifically focus on automotive paint drying systems remains scarce. This is attributed to the inherent complexity of these systems, which are characterized by a multitude of time-varying parameters and fluctuating energy consumption patterns influenced by factors such as ambient temperature and humidity, paint viscosity, conveyor speed, and airflow dynamics. Furthermore, the highly dynamic nature of the drying process, with frequent shifts in operating parameters based on production demands, makes it particularly challenging to establish a stable baseline for energy consumption. Therefore, it entails a tailored systematic methodology for effective anomaly detection and diagnosis of energy consumption within the paint-drying system, which must be capable of discerning genuine anomalies from intrinsic process variability and adapting to dynamic operating conditions.

To address this need, this study proposed a methodology (eAnoD) for the real-time energy consumption anomaly detection and diagnosis (RECADD) of paint-drying systems in automotive manufacturing. This method incorporated a specialized feature extraction technique that synergizes a temporal convolutional network (TCN) and graph attention network (GAT). This TCN-GAT architecture effectively distilled temporal and spatial characteristics from diverse data domains, including energy metrics, equipment parameters, production states, and environmental conditions. A real-time anomaly detection model built on a hybrid backpropagation neural network (BPNN) and variational autoencoder (VAE) framework utilized the extracted spatiotemporal features to generate an energy consumption anomaly index. Additionally, eAnoD introduced a hierarchical anomaly grading approach for both the entire drying system and its constituent subsystems to facilitate root cause analysis (RCA). This approach integrated combined weighting techniques with cloud modeling to enhance the granularity and accuracy of the diagnostic processes. Finally, the effectiveness and practicality of the proposed methodology were validated through a real-world case study conducted in a paint shop that produced a new-energy vehicle. This study provides a vivid demonstration and guideline for practical implementation and academic research.

The main contributions of this work are threefold.

(1) A systematic methodology eAnoD was developed for automotive paint drying systems to effectively identify and evaluate abnormal energy consumption patterns in real-time to minimize energy waste and enhance operational safety.

(2) A TCN-GAT architecture was designed to extract sophisticated spatiotemporal features from multidomain data, in conjunction with a hybrid neural network that combined BPNN and VAE to achieve accurate and robust anomaly detection.

(3) A hierarchical energy consumption anomaly grading method that employed combination weighting and cloud model to enable a nuanced assessment of anomaly severity and facilitate targeted maintenance and proactive risk prevention.

The remainder of this paper is organized as follows. Section II reviews the existing literature on energy consumption anomaly detection and diagnostic methods in an industrial context. Section III provides a detailed description of the proposed methodology using theoretical expressions. Section IV presents a real-life case study and discusses the empirical results. Finally, Section V concludes the paper, summarizes the key findings, and suggests potential avenues for future research.

2. Literature review

This section reviews the literature on anomaly detection and diagnostic approaches for energy management in industry.

2.1. Anomaly detection

Anomaly detection plays a crucial role in enhancing energy efficiency and alleviating energy waste in industrial processes [9]. Traditional methods that often rely on expert experience and manually defined thresholds for anomaly identification are increasingly being superseded by sophisticated algorithmic approaches. Yin et al. [10] introduced connectivity- and distance-based methods with dynamic thresholds to detect abnormal energy consumption during aluminum extrusion. Jin et al. [11] developed a spatial clustering method based on density for accurate anomaly detection in the steel production process by considering both energy data characteristics and manufacturing signals. S ndergaard et al. [12] proposed a continuous fault detection framework encompassing Chernoff bounds to minimize energy waste in district heating systems. However, these methods typically require extensive domain expertise for parameter tuning and struggle to adapt to the dynamic and complex nature of modern industrial processes. Furthermore, manual feature engineering using these approaches can be labor-intensive and may not effectively capture the underlying patterns indicative of anomalies, leading to suboptimal accuracy and limited generalization capabilities.

In contrast, deep learning techniques offer a powerful alternative that can automate feature extraction and learn complex non-linear relationships within data, thereby improving accuracy, adaptability, and generalizability across diverse industrial scenarios [13–15]. These techniques can be broadly categorized into prediction- and reconstruction-based approaches. Prediction-based methods learn from historical data to predict the expected behavior and flag deviations from these predictions as anomalies. For instance, Xu and Chen [16] developed an unsupervised neural network to generate prediction intervals for identifying abnormal building energy consumption. In contrast, reconstruction-based methods learn the latent data distribution of normal operations and detect anomalies as significant reconstruction errors. Zhao et al. [17] employed denoising autoencoders (DAE) to detect wind-turbine anomalies. Song et al. [18] designed a generative adversarial network (GAN) with multiple discriminators to monitor robot health based on the energy consumption. However, both these approaches face challenges. Prediction-based methods struggle with the scarcity of anomalous data for training, affecting the detection accuracy [19], whereas reconstruction-based methods

may exhibit sensitivity to anomalies during training and suffer from long data sequences, potentially corrupting the learned representations [20]. Furthermore, these methods often fall short in effectively capturing the intricate temporal dependencies and spatial correlations present in multivariate time-series data from complex industrial systems. This limitation can lead to suboptimal anomaly detection accuracy and hinder the interpretability of the results, making it difficult to identify the root causes of the detected anomalies.

Thus, this study proposed a novel approach that combined prediction- and reconstruction-based deep learning techniques with a specific feature extraction approach in a unified framework to derive spatiotemporal features from multidomain data and achieve accurate real-time anomaly detection. This hybrid approach leveraged the strengths of both the predictive and reconstructive approaches and enabled mutual complementation. Specifically, reconstruction-based methods are less reliant on labeled anomaly samples, whereas prediction-based methods excel at capturing long-term dependencies within the data. Combining these approaches can enhance both sensitivity to subtle anomalies and robustness of the detection process in the presence of noise and complex system dynamics.

2.2. Anomaly diagnosis

Anomaly diagnoses classify, assess, and localize abnormal conditions or events and determine their severity, priority, or root cause. The established methods include fuzzy theory, grey theory, artificial intelligence (AI) algorithms, and cloud models [21–23]. For instance, Xu et al. [24] developed an adaptive neurofuzzy inference system for anomaly diagnosis, achieving the precise detection of abnormal sound distances in the cooperative positioning of autonomous underwater vehicles. Although fuzzy theory effectively manages information with imprecise boundaries, it generally fails in scenarios involving both randomness and fuzziness, thereby compromising accuracy. To address this issue, Xu et al. [25] proposed a data mining approach based on multidimensional matrices to enhance the gray target theory for fault severity assessment in power grid equipment. AI algorithms excel in pattern recognition and predictive analytics but frequently suffer from limited interpretability [26].

In contrast, the cloud model theory offers a compelling alternative by combining the strengths of the fuzzy set theory and gray system theory to effectively address both randomness and fuzziness [27]. This approach allows for a more subtle representation of uncertainty, facilitating an accurate anomaly severity evaluation [28]. Cloud models have been successfully applied to grading and risk assessment in various scenarios. Guo et al. [29] combined cloud reasoning with expert scoring to assess natural gas pipeline risks, whereas Wu et al. [30] employed cloud models within a safety assessment framework to appraise island microgrid risks. Similarly, Liu et al. [31] introduced a Gaussian cloud-model-based health index to monitor hydropower unit condition shifts. Qi et al. [32] devised a combined weighting cloud model for the precise evaluation of coal and gas outburst risks. In contrast to traditional models, which require extensive datasets, cloud models deliver broad applicability and intuitive results with minimal data. Thus, this study adopted a cloud model paired with a combination weighting method to effectively evaluate the energy consumption anomaly severity for drying systems.

In addition to anomaly evaluation, RCA is critical for identifying the underlying triggers or factors of observed anomalies, thereby preventing recurrence and enhancing system reliability. RCA methodologies span from traditional approaches, such as the five Whys, fault tree analysis, and fishbone diagrams, to advanced techniques that leverage data-driven methods and AI [33,34]. Papa-

georgiou et al. [35] emphasized the increasing prominence of machine-learning-based RCA, employing decision trees and gradient boosting to identify defect causes in the pursuit of zero-defect manufacturing. Zhang et al. [36] introduced an autoencoder-based Shapley additive explanation framework for the RCA of operational faults in wind turbines by assigning importance values to features for specific predictions. Despite these advances, research that specifically addresses RCA for energy consumption anomalies in complex industrial systems remains limited, highlighting the need for further investigation. This study aimed to contribute to this field by designing a hierarchical anomaly grading approach for both the entire drying system and its subsystems to facilitate effective RCA.

2.3. Summary

The energy consumption of automotive paint-drying systems exhibits highly dynamic and irregular fluctuations influenced by a multitude of interacting factors, making it challenging to distinguish anomalies from normal operational variations. Moreover, the complex interplay and coupled relationships between multidomain data streams concerning temperature, pressure, gas flow, and other process parameters pose a significant challenge for conventional anomaly-detection methods. These methods generally struggle to effectively capture latent patterns and critical indicators of anomalous behavior in automotive paint drying systems. Thus, this study developed a hybrid anomaly-detection model that integrated predictive and reconstructive approaches to address the challenges of limited anomaly samples and the need for sensitive detection within complex and long-term dependencies. In addition, a hierarchical anomaly-grading method, leveraging combination weighting and cloud modeling, was devised to enable a nuanced assessment of anomaly severity regarding energy consumption for complex industrial systems and facilitate root cause localization and targeted interventions.

3. Methodology

Energy-consumption patterns are highly sensitive to operational anomalies in industrial systems that feature complex compositions, processes, and interactions. However, direct access to real-time anomaly information from local equipment is generally restricted by security and safety protocols. Therefore, the real-time monitoring of energy consumption, equipment operational parameters, environmental conditions, and production states facilitated by IoT technologies and data integration are crucial for detecting system abnormalities.

Thus, a systematic methodology, eAnoD, was proposed for RECADD based on multidomain data fusion. This approach comprised four key stages: data preprocessing, feature extraction, anomaly detection, and anomaly diagnosis. Initially, diverse energy sources were consolidated into a unified metric for comprehensive energy consumption assessment, followed by data cleaning procedures applied across all datasets. Subsequently, the architecture of TCN-GAT was designed to extract refined spatiotemporal features, which served as inputs for anomaly detection. A hybrid neural network integrating a BPNN and a VAE constructed a detection model and generated an anomaly index. Subsequently, a peaks-over-threshold (POT) method was applied to determine an adaptive anomaly threshold, enabling the precise identification of anomalies in conjunction with the anomaly index. Finally, based on the detected anomaly, a cloud model that incorporated qualitative and quantitative analyses was employed for anomaly grading. In addition, a dynamic combination weighting method was devel-

oped to assign context-aware weights to multiple evaluation indicators.

This methodology is expected to not only ensure accuracy, adaptability, and robustness in identifying and characterizing energy consumption anomalies but also offer potential generalizability to other industrial contexts characterized by complex systems and multisource data streams.

3.1. Data preprocessing

The energy consumed by drying systems mainly involves electricity and natural gas. To unify the evaluation criteria, these two types of energy should be jointly converted into standard coal according to the Comprehensive Energy Consumption Calculation Standard, as expressed as follows:

$$E = \sum_{i=1}^{N_e} (E_i \times k_i) \quad (1)$$

where E is the overall energy consumption in kgce (kilogram of coal equivalent), N_e is the source of energy, E_i is the actual consumption of the energy source, and k_i is the standard coal coefficient. In this study, the coefficients of electricity and natural gas were set to 0.1229 kgce·(kW·h)⁻¹ and 1.2143 kgce·m⁻³, respectively.

In addition to the energy data, other relevant factors, denoted by x , such as equipment operational parameters, environmental conditions, and production states, were also collected during the operation of the drying systems. Considering a raw data matrix comprising m time-series observations for each of n variables, this matrix can be represented as

$$\bar{\mathbf{X}} = [X_1, X_2, \dots, X_n] = \begin{bmatrix} x_{1,1} & x_{2,1} & \cdots & x_{n,1} \\ x_{1,2} & x_{2,2} & \cdots & x_{n,2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1,m} & x_{2,m} & \cdots & x_{n,m} \end{bmatrix} \in \mathbb{R}^{m \times n} \quad (2)$$

For cases where data were missing, a linear interpolation method was used to fill the gap. In addition, the min-max normalization technique was applied to rescale the variables to an interval between 0 and 1 to strengthen the model training performance.

A sliding time window with a width w was designed to enable the model to capture the temporal context of the data and retain memories of past information. Hence, each sample consisted of w consecutive data points, which provided the model with a comprehensive understanding of the temporal dynamics within the data, as depicted by

$$\bar{\mathbf{X}}' = \{x'_{1,st}, x'_{1,st+1}, \dots, x'_{1,st-w+1}; \dots; x'_{n,st}, x'_{n,st+1}, \dots, x'_{n,st-w+1}\} \quad (3)$$

$st = 1, 2, \dots, st - w + 1$

where st represents an arbitrary moment value within the sliding time window.

3.2. Spatial-temporal feature extraction

To ensure model performance, this study extracted both spatial and temporal features from the collected multidomain data. This approach was motivated by two key observations: ① the time-dependent nature of the data within a drying system, where the current value is closely related to previous values, and ② the spatial relationships among data points owing to the physical distribution of sensing devices, which are significant for mutual authentication. Model accuracy and interpretability are enhanced by these refined spatiotemporal features that influence anomaly identification and visualization, which aids in quick understanding. Moreover, these features capture potential spatial or temporal pat-

terns that are likely to be consistent across different datasets or periods, thereby improving their robustness and generalizability. Fig. 1 illustrates the specific process of spatiotemporal feature extraction for a drying system using TCN-GAT.

3.2.1. Temporal feature extraction

Several blocks of TCNs with various filter sizes were designed to extract temporal features from multidomain data. Each TCN comprised a three-layer temporal convolution module, with each layer consisting of two extended causal convolutions followed by weight normalization. The outputs were fed into the activation function of a rectified linear unit (ReLU). The following formula describes this process.

$$F(t) = \sum_{i=0}^{S_k-1} f_c(i) \cdot \mathbf{X}_{t-\varepsilon \cdot i} \quad (4)$$

where $f_c(i)$ is the convolution kernel, S_k is the size of the convolution kernel, ε is the expansion factor, and t is the index of current moment in the sequence. A Hadamard product was applied for this operation. The TCN transmitted the results to the posterior layer via the cross-layer residual connection R . Use N_r to denote the number of stacked residual blocks.

$$T_r = F(t) + \sum_{i=1}^{N_r-1} R(t_i, F_i) \quad (5)$$

where T_r represents the temporal feature extracted by each layer of the TCN and F is the dilated convolution operation. Explicitly, a dilated convolution introduces gaps into the kernel, which are filled with zeros between the non-zero elements. The temporal feature matrix $\mathbf{H}_{\text{temporal}}$ of the drying system combines those features. This design allows the network to have a wider receptive field without increasing the number of parameters or computational cost, and is particularly useful for capturing long-range dependencies in time-series data. The outcomes of each TCN were concatenated and then input into a pooling layer and a fully connected layer for feature fusion.

3.2.2. Spatial feature extraction

A GAT was constructed to model the spatial relationships between multiple factors within the system, leveraging its enhanced capacity and interpretability through the weighted importance of neighboring nodes. Specifically, each factor of the drying system, such as the gas flow rate, heat exchanger outlet temperature, and fan pressure difference, was represented as a node in the graph. A set of nodes was constructed as $H_s = \{\vec{h}_1, \vec{h}_2, \dots, \vec{h}_i\}$, $\vec{h}_i \in \mathbb{R}^D$, where h_i represents a node and D is the feature dimension for each node, as these factors are typically sequential in time. Hence, the size of D was related to the time series length of the factor inputted into the GAT model. The edges connecting these nodes represent the spatial proximity between factors. For example, adjacent factors in the same subsystem or in a close zone were connected by edges.

Additionally, the graph attention mechanism allowed each node to weigh the influence of its connected neighbors when computing its representation. These attention weights, learned during the training process, determined the relative importance of each neighbor's value in influencing the central node. It has been suggested that a global attention mode should be used without prior knowledge. To enhance the feature extraction performance, this study combined the multihead attention mechanism and multiple parallel score attention calculation processes, allowing the model to attend to different parts of the input in different ways. The characteristic matrix of node was obtained as follows:

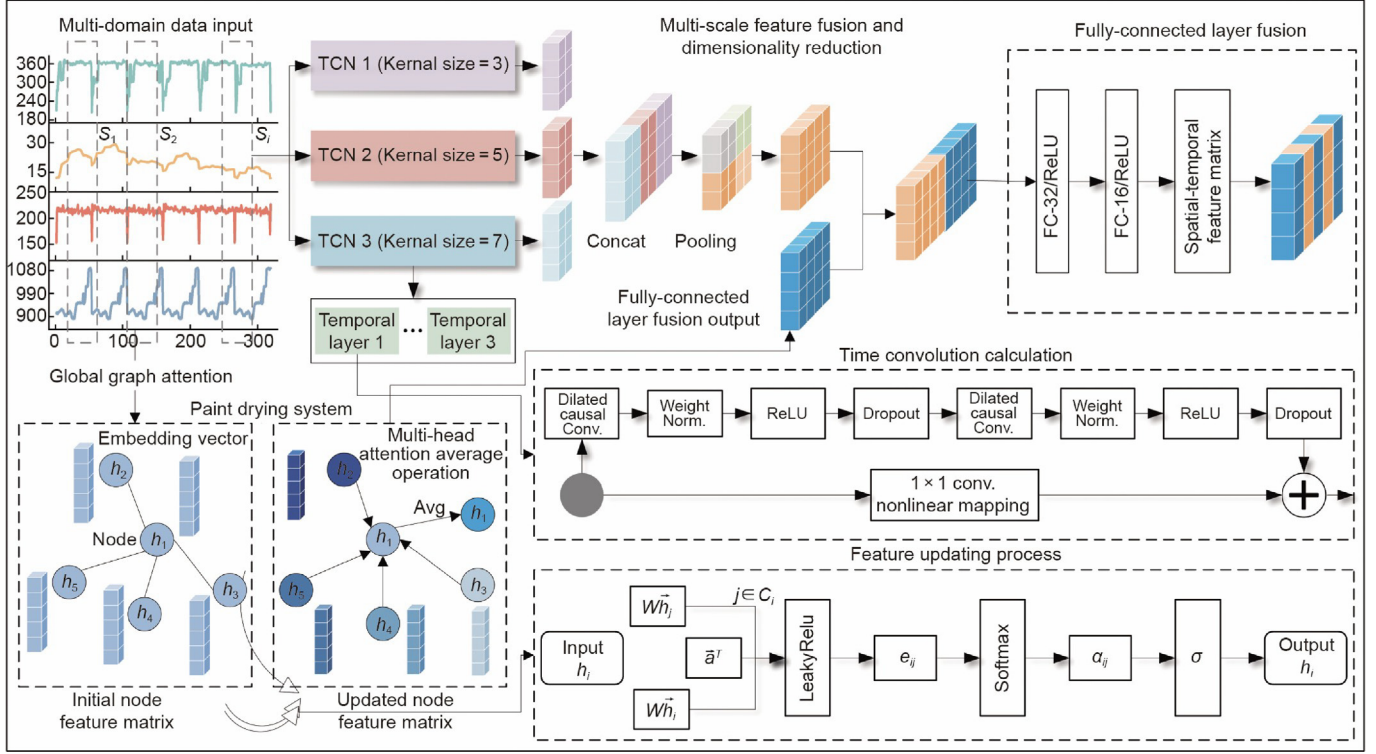


Fig. 1. The spatial-temporal feature extraction process for automotive drying systems using TCN-GAT. ReLU: rectified linear unit; h_i : the i th node in graph attention; $\vec{h}_i$: the embedding vector of the i th node; $\mathbf{W}h_i$: project the i th node's embedding vector via a weight matrix; $\mathbf{W}h_j$: project the j th node's embedding vector via a weight matrix; $\vec{a}^T$: the vector of learnable parameters with LeakyReLU; C_i : the entire node sets; e_{ij} : the features activated by LeakyReLU; α_{ij} : attention score; σ_s : the sigmoid activation function.

$$\vec{h}_i = \sigma_s \left(\frac{1}{K_a} \sum_{k=1}^{K_a} \sum_{j \in C_i} \alpha_{ij}^k \mathbf{W}^k \vec{h}_j \right) \quad (6)$$

where σ_s represents the sigmoid activation function and K_a is the number of heads for attention. The α_{ij} is the attention score that measures the degree of influence of node j on node i , which was obtained as follows:

$$e_{ij} = \text{LeakyReLU}(\vec{a}^T [\mathbf{W} \vec{h}_i || \mathbf{W} \vec{h}_j]) \quad (7)$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in C_i} \exp(e_{ik})} \quad (8)$$

$\mathbf{W} \in \mathbb{R}^{D' \times D}$ is a weight matrix, in which D' denotes the feature dimension of the output vector. $\vec{a}^T$ represents the vector of learnable parameters with LeakyReLU set as the activation function. The spatial feature matrix $\mathbf{H}_{\text{spatial}}$ of the drying system was obtained by amalgamating the output of each node.

The refined temporal and spatial features were inputted into the fully connected layers for fusion and dimensionality reduction. The process of securing the spatiotemporal characteristic matrix $\mathbf{H}$, which encapsulates the energy consumption information of the drying system operation, is presented as

$$\mathbf{H} = \text{FC}_2(\text{FC}_1(\mathbf{H}_{\text{temporal}}, \mathbf{H}_{\text{spatial}})) \quad (9)$$

where FC_1 and FC_2 represents two fully-connected neural networks.

The TCN-GAT model was trained in a supervised manner to predict the standard coal of total energy consumption during normal system operation. The target variable (standard coal) was calculated as the average energy consumption within a defined time window. Model training employed the mean squared error (MSE) loss function and Adam optimizer to minimize the prediction error. A learning-rate decay scheme was also implemented. The dataset

was partitioned as follows: 70% for training, 15% for validation, and 15% for testing. The training process involved early stopping based on the validation MSE to mitigate overfitting.

3.3. Energy consumption anomaly detection

3.3.1. Anomaly detection model

This study developed a specific anomaly detection model based on a hybrid neural network for accurate identification of energy consumption anomalies during the operation of drying systems, as shown in Fig. 2. The hybrid model leveraged the prediction model to learn the complex relationships between multiple key factors and energy consumption and effectively acquire normal behavior, and the reconstruction model to rebuild the input data based on this learned representation, with higher reconstruction errors indicating anomalies. This combination allowed the model to capture both complex relationships within the data and deviations from normality, leading to more accurate and robust anomaly detection.

The prediction network utilized a BPNN to model the nonlinear relationship between the extracted features and energy consumption data. The data supplied to the network are denoted as $\vec{x}$. The calculation process for each layer of the neural network is described in Eq. (10), where y_p is the overall energy consumption, $\mathbf{w}^T$ is the weight matrix, $\vec{b}$ is the bias vector, and $f_a(\cdot)$ is the activation function.

$$y_p = f_a(\mathbf{w}^T \vec{x} + \vec{b}) \quad (10)$$

To prevent overfitting, a dropout mechanism was incorporated into the first hidden layer. LeakyReLU was employed as an activation function to address the issue of neuronal death after nonlinear mapping. MSE was used to quantify the prediction error loss_{BP}, expressed as

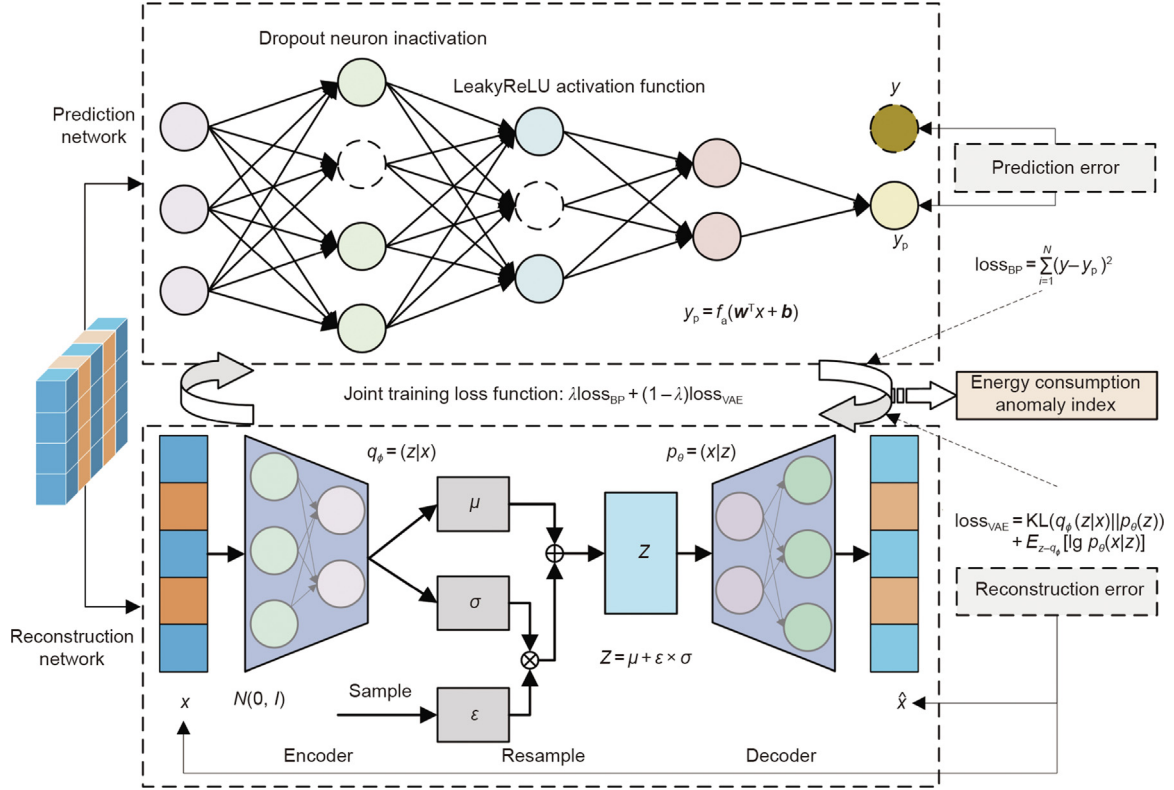


Fig. 2. The energy consumption anomaly detection model combining BPNN and VAE networks. λ : the relative weight of the loss function for prediction network; $N(0, I)$: sampling from a standard normal distribution; Z : samples of latent vectors; μ : the mean of samples; ϵ : a random number following a standard normal distribution of 0 and 1; σ : standard deviation of samples.

$$\text{loss}_{\text{BP}} = \frac{1}{N} \sum_{i=1}^N (y - y_p)^2 \quad (11)$$

where N is the number of predicted samples and y is the actual energy consumption of the drying system. The VAE served as the reconstruction network within the detection model and mainly consisted of an encoder and a decoder. In this study, the encoder and decoder were assumed to follow multi-variate Gaussian distributions in the latent space and original input space, respectively. The VAE detected anomalies by quantifying the dissimilarity between the original data matrix and its reconstruction. Thus, this study used the reconstruction probability loss_{VAE} to evaluate the anomaly, which was calculated as

$$\text{loss}_{\text{VAE}} = \text{KL}(q_\phi(z|x) || p_\theta(z)) + E_{z \sim q_\phi} [\lg p_\theta(x|z)] \quad (12)$$

The first part denotes the similarity between the approximate posterior distribution $p_\theta(z)$ and the prior distribution $q_\phi(z|x)$ of the hidden variable z , which was calculated using Kullback–Leibler (KL) divergence. $E_{z \sim q_\phi}$ denotes the reconstruction expectation that measures the decoder's ability to reconstruct the input. The second part denotes the similarity between the reconstructed and original data obtained using maximum likelihood estimation (MLE).

To integrate the advantages of prediction and reconstruction networks for realizing the synergistic optimization of network parameters and balancing the convergence states of both networks, the joint loss function of the anomaly-detection model was designed as follows:

$$\text{loss}_{\text{join}} = \lambda \text{loss}_{\text{BP}} + (1 - \lambda) \text{loss}_{\text{VAE}} \quad (13)$$

where λ is the relative weight of the loss function in regard to the BPNN. Moreover, VAE is favorable for learning a robust representation that is less sensitive to noise and irrelevant variations in the input data. In this case, the generalization and robustness of the entire anomaly detection model can be optimized.

3.3.2. Anomaly identification

This study introduced the POT model [37] to establish a threshold; an abnormal energy consumption index exceeding this threshold was regarded as an anomaly. The threshold was determined daily in response to the dynamic production plan and environmental changes in the drying system.

Specifically, a sequence of energy consumption anomaly indices is denoted by $S = \{s_1, s_2, \dots, s_t\}$, $t \in \mathbf{N}^+$. It was assumed that the part exceeding the threshold u followed the generalized Pareto distribution (GPD).

$$\bar{F}(s) = P(S - u > s | S > u) \sim_{\mu \rightarrow \tau} (1 + \frac{\xi s}{\beta})^{-\frac{1}{\beta}} \quad (14)$$

where u is the initial threshold, s is any value in the sequence, and $S - u$ is the indicator sequence exceeding the threshold u , which was empirically set to the high quantile. τ is the upper endpoint. In addition, ξ and β are the parameters of GPD, which were fitted using the MLE. Hence, the threshold for each day was obtained as follows:

$$u_t = u + \frac{\hat{\xi}}{\beta} \left(\left(\frac{q N_T}{N_u} \right)^{-\beta} - 1 \right) \quad (15)$$

where q denotes the probability of $s > u$ in relation to expected abnormal events. N_u represents the amount of data exceeding threshold u . q means the estimated probability of occurrence, and N_T denotes the total number of observations.

3.4. Energy consumption anomaly diagnosis

3.4.1. Combination weighting method

An accurate evaluation of energy consumption anomalies in automotive drying systems requires a multiindicator assessment, in which the scientific determination of indicator weights is crucial.

This study designed a combination weighting method based on the analytic hierarchy process (AHP) and the entropy weight method (EWM) to achieve weight allocation. This approach integrated expert knowledge with the inherent features of historical data to improve the reliability of anomaly level evaluations across a drying system and its subsystems. Note that the weight assigned to each indicator should reflect its contribution to the energy consumption at the respective system or subsystem levels for anomaly grading. For example, at the system level, indicator weights were determined based on the impact of the corresponding subsystem on the energy consumption of the overall system.

(1) **AHP.** The AHP incorporated expert knowledge, the impact of anomalies on the drying system, and its ability to determine indicator weights. First, judgment matrix $\mathbf{P}$ was constructed based on the defined anomaly level indices. The upper triangle of $\mathbf{P}$ reflected the pairwise relative importance of the indicators (scale: 1–10, with 1 indicating equal importance), whereas the lower triangle contained reciprocal values. The matrix is expressed as follows:

$$\mathbf{P}(p_{ij}) = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1l} \\ p_{21} & p_{22} & \cdots & p_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ p_{l1} & p_{l2} & \cdots & p_{ll} \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & p_{12} & \cdots & p_{1l} \\ \frac{1}{p_{21}} & 1 & \cdots & p_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{p_{l1}} & \frac{1}{p_{l2}} & \cdots & 1 \end{bmatrix} \quad (16)$$

where l is the number of indicators. The judgment matrix was subjected to a consistency test. The consistency index (CI) and consistency ratio (CR) were calculated using the following equations:

$$CI = \frac{\lambda_{\max} - O(\mathbf{P})}{O(\mathbf{P}) - 1} \quad (17)$$

$$CR = \frac{CI}{RI} \quad (18)$$

where $\lambda_{\max}$ is the maximum eigenvalue of the judgment matrix and $O(\mathbf{P})$ is the order of the judgment matrix. RI is the consistency index. If $CR < 0.1$, the judgment matrix passes the consistency test. Otherwise, the scoring must be adjusted and the matrix must be recalculated. Finally, the judgment matrix was normalized after passing the consistency test. This process yielded the subjective weight $\mathbf{W}^a$ of the anomaly indicators for drying systems.

(2) **EWM.** The EWM is an objective weighting method based on the degree of dispersion of various indicators in a dataset. The degree of dispersion was represented by the information entropy of each indicator. A larger information entropy value indicates a larger degree of dispersion and a smaller weight.

First, the original data of the indicators were standardized using the minimum–maximum technique, obtaining x'_{ij} . The proportion of each sample in the anomaly level indices was calculated using the following equation:

$$p_{ij} = \frac{x'_{ij}}{\sum_{i=1}^l x'_{ij}} \quad (19)$$

The entropy value e_j and information redundancy d_j of each anomaly level index were calculated as follows:

$$e_j = -\frac{1}{\ln(l)} \sum_{i=1}^l p_{ij} \ln(p_{ij}), j = 1, \dots, N_{\text{ind}} \quad (20)$$

$$d_j = 1 - e_j, j = 1, \dots, N_{\text{ind}} \quad (21)$$

N_{ind} means the number of indicators. Finally, the objective weight of each indicator pertaining to the entire system was determined as

$$W_j^{e1} = \frac{d_j}{\sum_{j=1}^{N_{\text{ind}}} d_j} \quad (22)$$

where d_j is information redundancy.

For subsystem-level indicators, connection entropy was used instead of information entropy, which is a combination of joint, mutual, and opposite entropies. Using the equation for calculating information entropy, the n -ary connection entropy of individual attribute was calculated as

$$H_j = -\frac{1}{(1/\ln(l) - p) \ln(N_{\text{ind}})} \left(\sum_{i=1}^{N_{\text{ind}}} f_{ij}(a) \ln f_{ij}(a) + \sum_{j=1}^{1/\ln(l)-2} \sum_{i=1}^{N_{\text{ind}}} f_{ij}(b) \ln f_{ij}(b) + \sum_{i=1}^{N_{\text{ind}}} f_{ij}(c) \ln f_{ij}(c) \right) \quad (23)$$

$$\text{where } f_{ij}(a) = \frac{a_{ij}}{\sum_{i=1}^{N_{\text{ind}}} a_{ij}}, f_{ij}(b) = \frac{b_{ij}}{\sum_{i=1}^{N_{\text{ind}}} b_{ij}}, f_{ij}(c) = \frac{c_{ij}}{\sum_{i=1}^{N_{\text{ind}}} c_{ij}}.$$

The p is the number of zero values among the joint, mutual, and opposite entropies. The objective weights for the subsystem-level indicators were computed as follows:

$$W_j^{e2} = \frac{1 - H_j}{\sum_{j=1}^{N_{\text{ind}}} (1 - H_j)} \quad (24)$$

(3) **Combination weighting.** This study applied an optimal combination weighting method to determine a comprehensive and accurate weight allocation scheme by considering both subjective and objective information.

First, the coefficients were assigned to the weight vectors derived using the AHP and EWM methods, generating a linear combination of two sets of weights.

$$\bar{\mathbf{W}} = \alpha_1 \mathbf{W}^a + \alpha_2 \mathbf{W}^e \quad (25)$$

where $\bar{\mathbf{W}}$ is a combined weight vector, α_1 is the coefficient of the weight vector of AHP $\mathbf{W}^a$, and α_2 is the coefficient of the weight vector of EWM $\mathbf{W}^e$.

Subsequently, the weight allocation coefficients were optimized to secure the optimal weight vector $\bar{\mathbf{W}}'$. A deviation minimization process was then applied to each weight vector as follows:

$$\min \sqrt{(\bar{\mathbf{W}}' - \mathbf{W}^a)^2 + (\bar{\mathbf{W}}' - \mathbf{W}^e)^2} \quad (26)$$

Finally, the linear coefficients were normalized to derive the final combination weights in accordance with the indicators.

3.4.2. Cloud model-based anomaly grading

To bridge the gap between qualitative anomaly assessments and quantitative index values for the energy consumption of a drying system, this study employed a cloud model for anomaly grading. The inherent data distribution patterns of the long-term operational data, rather than subjective rules, were used for classification. A backward cloud generator algorithm was developed to grade the anomalies for each key factor, which also served as an indicator. Moreover, it should be clarified that the ground truth for anomaly grading was derived by integrating expert knowledge with event logs recorded by the equipment.

(1) **Establish standard cloud.** Initially, multiple domains of sensing data were transformed into a numerical characteristic representation of the cloud model by calculating the cloud eigenvalues of the indicator data: the expected value Ex , the entropy En , and hyper-entropy He .

$$Ex = \frac{1}{N_c} \sum_{i=1}^{N_c} x_i \quad (27)$$

$$En = \sqrt{\frac{\pi}{2}} \times \frac{1}{N_c} \sum_{i=1}^{N_c} |x_i - \hat{Ex}| \quad (28)$$

$$He = \sqrt{S^2 - \hat{En}^2} \quad (29)$$

where N_c is the amount of sensing data sampled. Subsequently, based on the 3En principle of the cloud model and alarm thresholds, the grading standards for the drying system anomaly level assessment were established and are listed in Table 1.

With an assumption that the interval for grade g was $[x_{g,\min}, x_{g,\max}]$, Eqs. (30) and (32) were used to calculate the eigenvalues (Ex_g , En_g , He_g) of the cloud model corresponding to the g th interval.

$$Ex_g = \frac{x_{g,\min} + x_{g,\max}}{2} \quad (30)$$

$$En_g = \frac{x_{g,\max} - x_{g,\min}}{\sqrt{-8 \ln 0.5}} \quad (31)$$

$$He_g = \gamma En_g \quad (32)$$

In these equations, γ is typically set to 0.1. A standard cloud of indicators for anomaly evaluation was generated using a backward cloud generator.

(2) **Anomaly level grading.** This process began with the acquisition and normalization of the evaluation indicator data. Subsequently, the cloud model eigenvalues for each indicator were calculated using a backward cloud generator. For a subsystem layer comprising h indicators, the eigenvalues were determined as follows:

$$Ex = \sum_{j=1}^{N_{ind}} \bar{w}_j Ex_j \quad (33)$$

$$En = \sqrt{\sum_{j=1}^{N_{ind}} \bar{w}_j En_j^2} \quad (34)$$

$$He = \sum_{j=1}^{N_{ind}} \bar{w}_j He_j \quad (35)$$

The eigenvalues of each indicator were calculated iteratively. Upon completion, the membership degree of the indicator data was computed and a corresponding evaluation cloud chart was generated. The anomaly level of the drying system was then assessed by identifying the highest similarity between the evaluation and standard clouds. The following procedure details the process for obtaining cloud similarity.

Considering two clouds, $C_1(Ex_1, En_1, He_1)$ and $C_2(Ex_2, En_2, He_2)$, we let u_i represent the membership degree of a cloud droplet (x_i, μ_i) generated by the backward cloud generator of C_1 within cloud C_2 , which was computed as follows:

$$\mu_i = e^{-\frac{(x_i - Ex_2)^2}{2(En_2)^2}} \quad (36)$$

These steps were repeated N_{step} times, resulting in N_{step} membership degrees. The similarity δ between the two clouds was calculated as follows:

$$\delta = \frac{1}{N_{step}} \sum_{i=1}^{N_{step}} \mu_i' \quad (37)$$

To ascertain the validity of the observed improvements in anomaly grading, statistical significance tests were performed and a comprehensive error analysis was conducted. The anomaly grade of the overall system was used for a holistic assessment of the detected anomalies. Furthermore, anomaly localization was facilitated by comparing the identified anomaly levels across the subsystems of the drying system. Subsystems exhibiting higher severity levels were prioritized for further investigation, including a detailed analysis of their corresponding indicator data. This hierarchical approach enables the effective and efficient detection of energy consumption anomalies and their root causes, preventing escalation and reducing potential safety risks.

4. Case study

4.1. Experimental settings

To verify the efficacy of the proposed approaches for energy consumption anomaly detection and diagnosis, a real-world case study was conducted in the paint shop of a new-energy vehicle manufacturer in Chongqing, a major industrial hub in Southwest China. This study focused on developing a real-time anomaly detection service system for the electrophoretic drying process, which is a critical stage in automotive paint shops where vehicle bodies undergo protective coating. Proper drying after electrophoresis is essential for ensuring uniform paint film solidification and preventing defects such as uneven drying, orange peeling, and cracking. These defects compromise the protective quality of the coating and negatively affect the adhesion of subsequent paint layers. The service system not only detects anomalies across the entire process but also provides anomaly grading for individual subsystems once a system-level anomaly has been identified. The drying system was divided into several subsystems, including combustion heating, hot air circulation, fresh air heat exchange, and cooling systems. Fig. 3 shows the entire structure of the drying system with on-site scenes.

4.1.1. Dataset establishment

Data collection spanned from September 1, 2023 to November 30, 2023 at 15 min intervals. The data collected comprised multidomain factors related to energy consumption, equipment status, production states, and environmental conditions. These data were sourced from various IoT devices and enterprise information systems such as manufacturing operation management (MOM) systems. Table 2 lists some actual samples of each factor. The overall energy consumption was calculated by aggregating natural gas and electricity consumption data. This initial collection resulted in 8684 data points. Data interpolation generated a dataset comprising 8736 samples across 21 indicators (8736×21). To facilitate temporal analysis, the dataset was transformed using a sliding window technique with a window size of 15 and a step size of 1, resulting in a final dataset dimension of $8722 \times 15 \times 21$.

Based on historical records, the collected dataset contained the following abnormal events during the operation of the

Table 1
Grading standard of anomaly evaluation index for drying system.

Condition	Slight anomaly	Moderate anomaly	Major anomaly	Critical anomaly
$Ex + En < A_{\max} < Ex + 2En$	$(Ex + En, A_{\max})$	$(A_{\max}, Ex + 2En)$	$(Ex + 2En, A_{\max} + En)$	$(A_{\max} + En, Ex + 3En)$
$Ex + 2En < A_{\max} < Ex + 3En$	$(Ex + En, Ex + 2En)$	$(Ex + 2En, A_{\max})$	$(A_{\max}, Ex + 3En)$	$(Ex + 3En, A_{\max} + En)$
$A_{\max} > Ex + 3En$	$(Ex + En, Ex + 2En)$	$(Ex + 2En, Ex + 3En)$	$(Ex + 3En, A_{\max})$	$(A_{\max}, A_{\max} + En)$

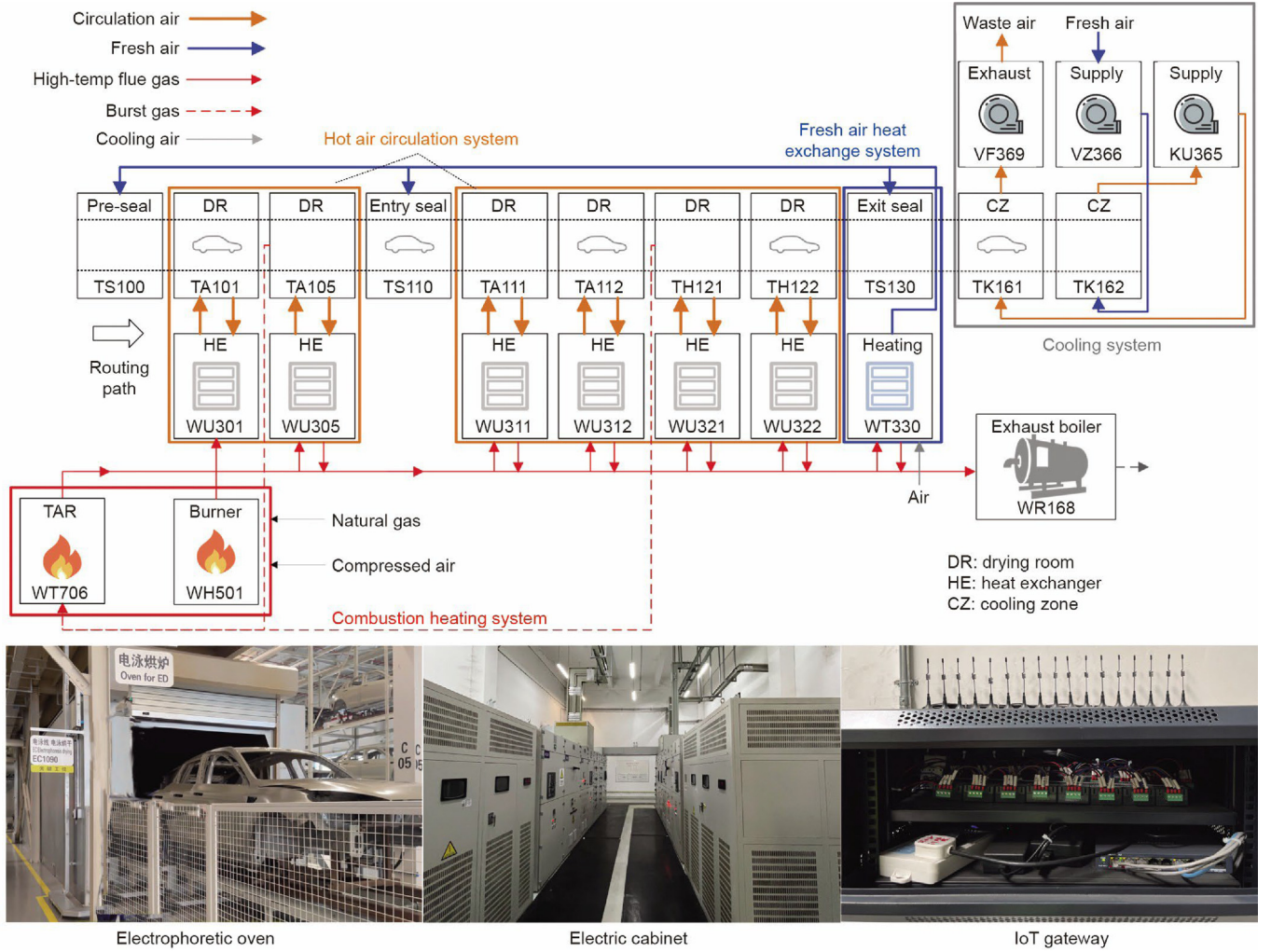


Fig. 3. The structure of automotive paint drying system with actual scenes. TAR: thermal-activated recirculation in waste gas incineration system.

drying system. ① On November 20 and November 22, the heat exchanger (WU301) failed to reach its preset temperature within the allotted time, preventing the drying system from initiating heating in the insulation zone. This malfunction caused deviations in the energy consumption curve. ② On November 26, a blockage in the burner's control valve led to unstable gas pressure. To compensate for this, the temperature control system increased the opening of the natural gas valve, resulting in an elevated system energy consumption. ③ On November 30, a prolonged failure of the heat exchanger (WU301) fan caused a decrease in system energy consumption. In this case, the data from September 1 to November 15 were used as the normal operation data for the training set.

The prediction model within the hybrid anomaly detection method was trained on data automatically labeled by the average energy consumption within a synchronized time window. The data from November 16th to 30th were used as the test set, with anomaly labels assigned manually based on expert knowledge and event logs. This test set was further divided into two equal-sized datasets to evaluate the robustness of the model under different operational scenarios (Table 3).

4.1.2. Model settings

The specific model parameters of TCN-GAT for spatiotemporal feature extraction are listed in Table 4. The model was trained

using an Adam optimizer with a learning rate of 0.001, regularization rate of 0.2, and batch size of 32.

For the hybrid anomaly-detection model, the BPNN consisted of three hidden layers with neuron counts of 32, 64, and 32, whereas the VAE was configured with a latent variable dimension of 4, resampling size of 10, and output matrix dimension of 8. Both the BPNN and VAE were trained using an Adam optimizer with a learning rate of 0.001 and a batch size of 32. The weighting factor λ for the prediction loss in the total loss function was set to 0.7.

The experimental setup utilized a software environment involving Python 3.9, PyTorch 1.12 deep-learning framework, and CUDA 11.3. The hardware platform comprised a Dell Precision 3640 workstation (Dell, China) running Windows 10, equipped with an Intel Core i9-10900 CPU (Intel, USA), an NVIDIA RTX 2080 Ti GPU (NVIDIA, China), and 64 GB of RAM (Micron, USA).

4.2. Experimental results and discussion

4.2.1. Anomaly detection

This section demonstrates the performance of the proposed methodology for the RECADD of an automotive paint drying system. When the system operated normally, the calculated energy consumption anomaly index remained within a narrow band, consistently below the threshold determined by the POT fitting. However, when the system deviated from its normal operating

Table 2

The sample data of drying system for anomaly detection and diagnosis.

System	Equipment	Collected parameter	Unit	2023-09-01 00:00	2023-09-01 00:15	2023-11-30 23:45
Combustion heating system	Gas meter	Natural gas flow-rate V_{gas}	$\text{m}^3 \cdot \text{h}^{-1}$	0.4500	0.4600	0.4000
		Natural gas pressure P_{gas}	bar	206.7546	199.2641	196.7790
		Compressed air flow V_{air}	$\text{m}^3 \cdot \text{h}^{-1}$	53.1501	53.6131	53.0472
		Compressed air pressure P_{air}	bar	2842.3712	2821.3573	2816.6770
Low-temp zone of hot air circulation system	Heat exchanger WU301	Outlet temperature T_{WU301}	$^{\circ}\text{C}$	87.7268	84.3041	92.4576
		Fan pressure difference P_{WU301}	Pa	985.7992	941.4691	945.8357
		Outlet temperature T_{WU305}	$^{\circ}\text{C}$	91.8505	88.0155	95.1895
	Heat exchanger WU305	Fan pressure difference P_{WU305}	Pa	871.3763	827.7887	799.8168
		Outlet temperature T_{WU311}	$^{\circ}\text{C}$	202.4742	194.2268	201.5335
		Fan pressure difference P_{WU311}	Pa	1139.1655	1087.9481	1050.1522
High-temp and insulation zone of hot air circulation system	Heat exchanger WU311	Outlet temperature T_{WU312}	$^{\circ}\text{C}$	209.7732	201.1959	199.8039
		Fan pressure difference P_{WU312}	Pa	859.7018	901.0312	852.8688
		Outlet temperature T_{WU321}	$^{\circ}\text{C}$	205.8415	200.3581	201.0850
	Heat exchanger WU312	Fan pressure difference P_{WU321}	Pa	846.2278	841.8042	854.8539
		Outlet temperature T_{WU322}	$^{\circ}\text{C}$	210.7216	201.6495	200.9524
		Fan pressure difference P_{WU322}	Pa	995.0804	950.4124	912.7543
	Heat exchanger WU321	Outlet temperature T_{WU330}	$^{\circ}\text{C}$	192.4147	184.2268	167.1398
		Fan pressure difference P_{WU330}	Pa	1989.4853	1974.5362	1946.1635
		Environmental temperature T_0	$^{\circ}\text{C}$	28.1633	28.1013	17.1458
	Heat exchanger WU322	Drying system yield Q	piece	7.0000	8.0000	2.0000
		Holistic energy consumption E	kgce	92.1247	88.5453	36.8854

Table 3

The division of datasets.

Dataset	Normal data (pieces)	Data proportion	Abnormal data (pieces)	Data proportion
Training set	7378	100%	0	0
Test set 1	672	97.0%	20	3.0%
Test set 2	672	93.6%	43	6.4%

conditions, the anomaly index increased significantly, exceeding the established threshold, as shown in Fig. 4.

Table 4

The settings of the TCN-GAT model's hyper-parameters.

Component	Hyper-parameter	Value
TCN	No. of TCN networks	3
	No. of time conv. modules	3
	Size of conv. kernel	3, 5, 7
	No. of hidden layer neurons	[32, 64, 32]
Temporal feature fusion network	Activation function	ReLU
	Average pooling layer window size	3×3
	Step	3
	Fully-connected layer	[32, 16]
GAT	No. of nodes	21
	No. of attention heads	4
	Activation function	LeakyReLU
	Node feature dimension	15
Spatial-temporal feature fusion network	Fully-connected layer	[32, 16, 8]

In general, the drying system required 2.5 h to reach the predefined temperature after startup. A longer time taken to warm the drying system or a sudden increase in temperature was considered an abnormality, thus triggering alerts. Notably, the model successfully identified these anomalies at 11:45 on November 20 and 07:15 on November 22, which aligned with the system inspection logs.

Fig. 5 shows the model performance over two rounds of continuous production, each lasting for several days without interruption. The results illustrated the effectiveness of the model in identifying anomalies, even amidst the regular fluctuations inherent in continuous operations. The model detected anomalies at 02:30 on November 26, 02:00 on November 27, and 04:45 on November 30, accurately reflecting incidents recorded in the system inspection logs.

These findings demonstrated the efficacy of the proposed anomaly-detection model in identifying deviations from normal energy consumption patterns across diverse operational scenarios. Furthermore, to rigorously evaluate the proposed methods, a robustness test and a comparative analysis were conducted.

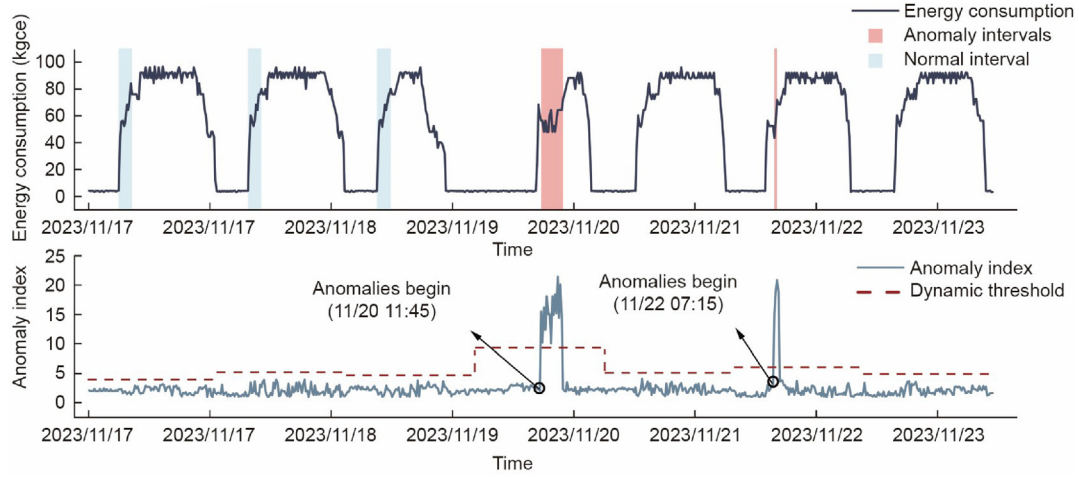


Fig. 4. The results of anomaly detection based on the test set 1.

(1) **Robustness test.** The robustness of the model to variations in the training data volume was assessed by altering the training set size (60%, 70%, 80%, 90%, and 100%). The experimental results are listed in Table 5.

The results show that even with a 40% reduction in training data (60% of the original set), the model maintained an F1-score > 0.85 across the two test datasets. This result indicates the ability of the model to maintain reliable performance despite significant reductions in training data, which is beneficial for practical implementation.

(2) **Comparative analysis.** The proposed methodology, eAnoD, was compared against four benchmark anomaly-detection models: VAE, GAN, one-class convolutional neural network (OC-CNN), and long short-term memory (LSTM) network. As presented in Table 6, this approach outperformed all benchmark models in terms of precision ratio, recall, and F1-score on Test Set 1.

Specifically, eAnoD achieved a 3% higher F1-score compared to the best-performing benchmark method, namely, OC-CNN. This enhanced performance was attributed to the incorporation of the TCN and GAT into the model, enabling the effective extraction of spatiotemporal features from multidomain data. Although OC-CNN demonstrated a marginally higher accuracy in Test Set 2, its significantly lower recall suggested a tendency to misclassify anomalies as normal, which is a critical drawback in practical applications. This discrepancy could be attributed to two factors. First, eAnoD had reduced bias; thus, it identified more anomalies at the expense of

some false positives. Second, the subtle nature of the anomalies in Test Set 2 posed a challenge for the detection capabilities of the OC-CNN, leading to lower recall. Considering the potential consequences of missed detections against misjudgments of energy consumption anomalies in drying systems, prioritizing recall over accuracy over precision is crucial. In addition, the superior F1-score of the proposed method further proved its balanced performance in minimizing both false positives and false negatives.

However, the GAN performed consistently poorly, with less than 90% performance for all metrics in both datasets, owing to its inability to handle discrete and high-dimensional data. In addition, LSTM performed slightly worse than OC-CNN because the sequential dependencies in the data were weak. Therefore, the results indicate that the improved anomaly-detection capability via eAnoD effectively resulted in accurate and robust identification of potential faults, enabling proactive maintenance and reducing downtime.

4.2.2. Anomaly diagnosis

Anomaly levels were determined for the entire drying system and its constituent subsystems using the combination weighting and cloud modeling. Each subsystem's contribution to the overall system-level anomaly was weighted to reflect its influence on the energy consumption. The system-level anomaly grade was used to trigger alerts, whereas the subsystem-level grades facilitated RCA, enabling a rapid response and targeted action.

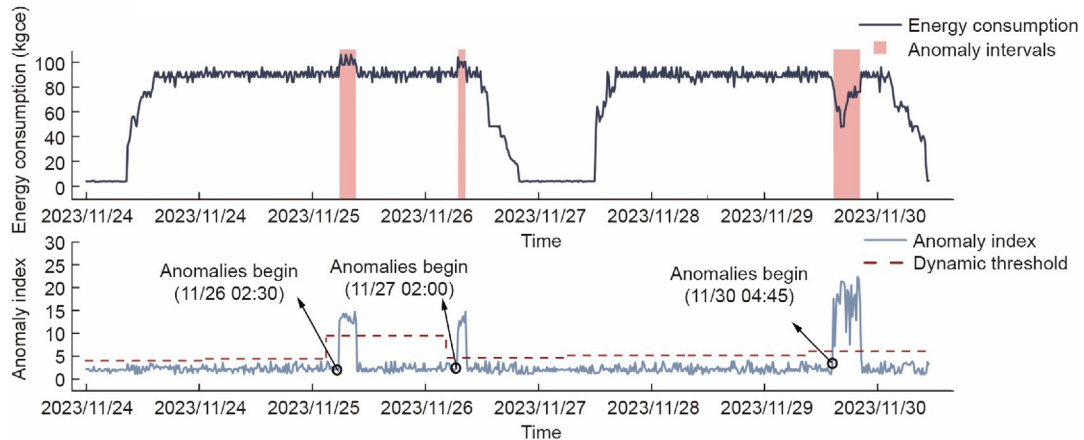


Fig. 5. The results of anomaly detection based on the test set 2.

Table 5

The anomaly detection results with diverse sizes of training dataset.

Dataset	Proportion of training dataset	Precision ratio	Recall ratio	F1-score
Test set 1	60%	0.8840	0.8910	0.8875
	70%	0.8915	0.8961	0.8938
	80%	0.9215	0.8984	0.9098
	90%	0.9417	0.9242	0.9329
	100%	0.9764	0.9522	0.9641
Test set 2	60%	0.8474	0.8714	0.8592
	70%	0.8855	0.8925	0.8890
	80%	0.9018	0.9284	0.9149
	90%	0.9178	0.9202	0.9190
	100%	0.9512	0.9635	0.9573

Table 6

The comparative results of anomaly detection methods for the automotive drying system.

Dataset	Model	Index		
		Precision ratio	Recall ratio	F1-score
Test set 1	OC-CNN	0.9415	0.9338	0.9376
	LSTM	0.9248	0.9214	0.9231
	VAE	0.9012	0.9133	0.9072
	GAN	0.8807	0.8835	0.8821
	eAnoD	0.9764	0.9522	0.9641
Test set 2	OC-CNN	0.9608	0.9211	0.9405
	LSTM	0.9555	0.8791	0.9157
	VAE	0.8998	0.9024	0.9011
	GAN	0.8798	0.8994	0.8895
	eAnoD	0.9512	0.9635	0.9573

Standard cloud establishment.

(1) Combination weighting: The AHP facilitated subjective weighting. A panel of nine experts from academia, equipment manufacturers, and the case company was assembled to evaluate the relative importance of the indicators for diagnosing energy consumption anomalies in the drying system. The experts considered the potential impact of the anomalies within each subsystem and their associated maintainability when assigning scores. The objective weights were determined using the EWM. The final indicator weights for each subsystem were derived by combining the subjective and objective weights, as presented in Table 7.

The results indicated the critical influence of the combustion heating system on the overall drying system performance. As the primary heat source, its operational state directly affected the performance of the subsystems responsible for delivering the heated air. This also highlights the crucial role of temperature control in hot-air circulation systems. The combination weighting approach appropriately prioritized temperature over the fan pressure difference, revealing tighter temperature control requirements and greater operational variability permitted for the fan pressure difference. This validated the effectiveness of the weighting method in accurately reflecting the relative contributions of different indicators to anomaly grading.

(2) Standard cloud: By leveraging the cloud model generator and historical operational data of the drying system, the threshold intervals for each evaluation indicator were established across the four anomaly levels. These thresholds were determined in accordance with the commonly observed abnormal conditions for each indicator, as listed in Table 8.

Subsequently, with the assistance of the settled index weights, the cloud model generator derived the eigenvalues for each anomaly level, as presented in Table 9. Fig. 6 shows the resulting standard clouds for the overall drying system and its subsystems, which provided a comprehensive representation of the operational states under various anomalous conditions.

Evaluation process. The efficacy of the proposed anomaly-grading model was illustrated through an analysis of an abnormal

event recorded at 02:30 on November 26, 2023. The cloud eigenvalues of the abnormal samples were calculated using the backward cloud generator algorithm. These eigenvalues were then weighted according to a previously defined scheme to derive the comprehensive cloud eigenvalues for each subsystem and the entire drying system, as presented in Table 10.

Based on the cloud eigenvalues, evaluation cloud charts for each subsystem and drying system were generated, as shown in Fig. 6. The resemblance between these evaluation and standard clouds for each anomaly level was quantified using the cloud similarity metric presented in Table 11 for both the overall system and individual subsystems.

Based on the abovementioned results, the findings of this study are summarized as follows.

(1) The evaluation cloud of the combustion heating system, as shown in Fig. 6(a), exhibited the highest similarity (0.8791) with the “Major Anomaly” standard cloud, indicating a significant deviation from normal operating conditions. This suggests the need for prompt intervention, such as postproduction inspection and maintenance, to address potential issues.

(2) As shown in Fig. 6(b), the evaluation cloud of the low-temperature zone of the hot-air circulation system closely aligns with the standard cloud of the general anomaly, with a cloud similarity of 0.7585, indicating a general anomaly level. This suggests a moderate deviation, requiring enhanced monitoring of key parameters and potentially increased equipment inspection frequency.

(3) The evaluation clouds of the high-temperature and insulation zones of the hot-air circulation system and fresh-air heat exchange system, as shown in Figs. 6(c) and (d), are in close proximity to the slightly abnormal standard cloud, with cloud similarities of 0.7423 and 0.6241, respectively. This implies a minor deviation from normal operation that entails continued observation of anomaly indices but not immediate intervention.

(4) Fig. 6(e) illustrates that the evaluation cloud of the drying system resembled the standard cloud of the general anomaly, with a cloud similarity value of 0.8747. This signifies a moderate

Table 7

The weights allocation of subsystem in the drying system for anomaly grading.

Subsystem	AHP		EWM		Combination	
	Subsystem weight	Indicator weight	Subsystem weight	Indicator weight	Subsystem weight	Indicator weight
Combustion heating system	0.320	[0.504, 0.124, 0.304, 0.068]	0.314	[0.430, 0.134, 0.342, 0.094]	0.325	[0.460, 0.130, 0.327, 0.083]
Low-temp zone of hot air circulation system	0.220	[0.440, 0.296, 0.160, 0.104]	0.244	[0.277, 0.262, 0.281, 0.180]	0.200	[0.277, 0.262, 0.281, 0.180]
High-temp and heat preservation zone of hot air circulation system	0.281	[0.238, 0.227, 0.129, 0.130, 0.087, 0.086, 0.053, 0.050]	0.253	[0.128, 0.095, 0.119, 0.147, 0.131, 0.164, 0.118, 0.098]	0.305	[0.221, 0.207, 0.127, 0.133, 0.094, 0.098, 0.063, 0.057]
Fresh air heat exchange system	0.179	[0.571, 0.429]	0.189	[0.577, 0.423]	0.170	[0.651, 0.349]

Table 8

The abnormal level grading of evaluation indicators in the drying system.

Evaluation index	Abnormal level			
	Slightly anomaly	General anomaly	Larger anomaly	Vital anomaly
P_{gas} bar	(0.482, 0.500)	(0.500, 0.532)	(0.532, 0.549)	(0.549, 0.582)
Q_{gas} m ³ ·h ⁻¹	(208, 215)	(215, 220)	(220, 224)	(224, 228)
P_{air} bar	(53.729, 54.136)	(54.136, 54.544)	(54.544, 55)	(55, 55.408)
Q_{air} m ³ ·h ⁻¹	(3175.711, 3470.623)	(3470.623, 3600)	(3600, 3765.541)	(3765.541, 3894.911)
T_{WU301} °C	(94.714, 96)	(96, 99.117)	(99.117, 100.402)	(100.402, 103.518)
P_{WU301} Pa	(984.728, 998.792)	(998.792, 1012.857)	(1012.857, 1020)	(1020, 1034.064)
T_{WU305} °C	(108.385, 110)	(110, 111.270)	(111.270, 112.885)	(112.885, 114.155)
P_{WU305} Pa	(907.098, 910.998)	(910.998, 914.898)	(914.898, 920)	(920, 923.900)
T_{WU311} °C	(203.593, 205.000)	(205.000, 206.885)	(206.885, 208.292)	(208.292, 210.178)
P_{WU311} Pa	(898.685, 903.033)	(903.033, 907.382)	(907.382, 910)	(910, 914.348)
T_{WU312} °C	(201.803, 204.327)	(204.327, 205)	(205, 206.851)	(206.851, 207.524)
P_{WU312} Pa	(825.470, 830.357)	(830.357, 835.244)	(835.244, 840)	(840, 844.886)
T_{WU321} °C	(204.231, 208)	(208, 210.939)	(210.939, 211.35)	(211.354, 214.29)
P_{WU321} Pa	(811.377, 815.816)	(815.816, 820.255)	(820.255, 830)	(830, 834.438)
T_{WU322} °C	(202.665, 205.724)	(205.724, 208.782)	(208.782, 210)	(210, 213.058)
P_{WU322} Pa	(785.036, 788.577)	(788.577, 792.118)	(792.118, 800)	(800, 803.541)
$P_{\text{fersh air}}$ Pa	(1953.799, 1957.609)	(1957.609, 1961.419)	(1961.419, 1980)	(1980, 1983.809)
$T_{\text{fersh air}}$ °C	(174.418, 186.725)	(186.725, 192)	(192, 199.032)	(199.032, 204.307)

Table 9

The eigenvalues of the drying system and its subsystems.

System and subsystem	Abnormal level	Ex	En	He
Combustion heating system	Slight anomaly	0.107	0.039	0.004
	Moderate anomaly	0.365	0.054	0.005
	Major anomaly	0.621	0.038	0.004
	Critical anomaly	0.864	0.049	0.005
Low-temp zone of hot air circulation system	Slight anomaly	0.118	0.042	0.004
	Moderate anomaly	0.373	0.050	0.005
	Major anomaly	0.616	0.038	0.004
	Critical anomaly	0.861	0.050	0.005
High-temp and insulation zone of hot air	Slight anomaly	0.140	0.050	0.005
	Moderate anomaly	0.401	0.044	0.004
	Major anomaly	0.639	0.042	0.004
	Critical anomaly	0.878	0.044	0.004
Fresh air heat exchange system	Slight anomaly	0.113	0.041	0.004
	Moderate anomaly	0.299	0.026	0.003
	Major anomaly	0.613	0.087	0.009
	Critical anomaly	0.928	0.026	0.003
Drying system	Slight anomaly	0.125	0.045	0.004
	Moderate anomaly	0.371	0.044	0.004
	Major anomaly	0.625	0.048	0.005
	Critical anomaly	0.873	0.046	0.005

system-wide anomaly, highlighting the need for the enhanced maintenance of key components within the affected subsystems.

The identified anomaly, which was attributed to a blockage in the burner's regulating valve, agreed with the analysis results. This blockage, which caused unstable gas pressure, prompted the temperature control system to increase the natural gas flow, resulting in elevated consumption and the observed energy anomaly. The subsystem-level evaluations further validated this finding. The

combustion heating system exhibited the most significant deviation, clearly indicating its root cause. Other connected subsystems exhibited "Moderate" or "Slight" anomalies caused by the increased combustion. The overall "Moderate Anomaly" assessment accurately reflected the severity of this common equipment issue, which is unlikely to result in immediate critical failures.

The consistency between the anomaly grading results and the actual operational log demonstrates the effectiveness and practi-

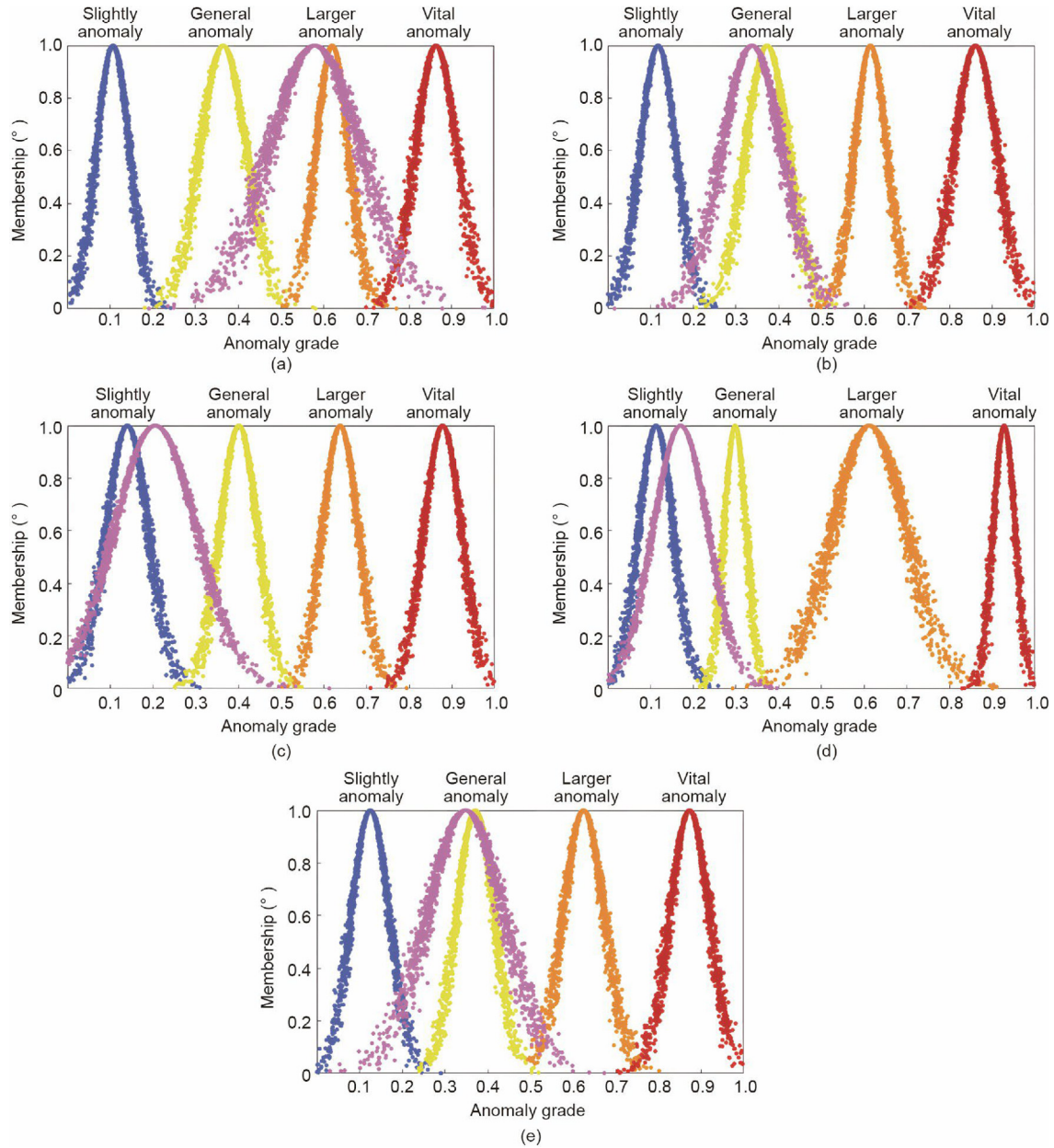


Fig. 6. The evaluation clouds of the drying system and its subsystems for anomaly grading: (a) combustion heating system; (b) low-temp zone of hot air circulation system; (c) high-temp and insulation zone of hot air circulation system; (d) fresh air heat exchange system; (e) drying system.

Table 10

The cloud eigenvalues of drying system and its subsystems.

System and subsystem	Cloud eigenvalues		
	Ex	En	He
Combustion heating system	0.580	0.107	0.011
Low-temp zone of hot air circulation system	0.338	0.070	0.007
High-temp and insulation zone of hot air circulation system	0.206	0.097	0.005
Fresh air heat exchange system	0.171	0.065	0.003
Drying system	0.348	0.089	0.009

quality of the proposed methodology. This approach provides a robust framework for assessing the severity of anomalies within a drying system, enabling timely and targeted interventions to ensure system stability and optimize energy efficiency.

5. Conclusions

This study developed eAnoD, a systematic methodology for RECADD in paint drying systems, to improve energy efficiency and mitigate risks in automotive manufacturing. This methodology comprised three key components: a TCN-GAT architecture for extracting informative spatiotemporal features from multidomain data; a hybrid BPNN-VAE model for accurate and robust anomaly detection; and a cloud model with a combination weighting scheme for grading anomalies at both system and subsystem levels. This hierarchical grading approach facilitates RCA and enables effective predictive maintenance. A real-world case study was conducted in a new-energy vehicle factory to validate the effectiveness and superiority of the proposed method. This methodology has the potential to be generalized to other industrial RECADD applications characterized by complex processes and multidomain data fusion. This study can provide practical

Table 11

Cloud similarity evaluation of abnormal level for drying system and its subsystems.

System and subsystem	Cloud similarity			
	Slight anomaly	Moderate anomaly	Major anomaly	Critical anomaly
Combustion heating system	0.0001	0.1788	0.8791	0.0519
Low-temp zone of hot air circulation system	0.0206	0.7585	0.0018	0.0001
High-temp and insulation zone of hot air circulation system	0.7423	0.1744	0.0001	0.0001
Fresh air heat exchange system	0.6241	0.1792	0.0001	0.0001
Drying system	0.0676	0.8747	0.0196	0.0001

guidelines for practitioners facing similar challenges and contribute a novel systematic methodology to the field of RECADD.

However, this study has certain limitations. The collected data, spanning a specific timeframe, yielded a limited set of anomalous samples that were potentially insufficient to represent the full spectrum of anomalous behavior. Furthermore, the evaluation focused solely on a paint-drying system for energy monitoring, although the proposed methodology has the potential for broader applications. Hence, future research should address these limitations by expanding anomaly detection to include the entire paint shop and other manufacturing workshops. It would be beneficial to pursue further practical applications to demonstrate the generalizability of the proposed methodology and its contribution to broader big data analytics initiatives within the manufacturing domain. Additionally, it is necessary to explore more advanced neural network architectures to enhance the performance across diverse anomaly types.

CRediT authorship contribution statement

Wei Wu: Writing – review & editing, Writing – original draft, Project administration. **Congbo Li:** Supervision, Project administration. **Youhong Zhang:** Investigation, Data curation. **Hewang Zhai:** Writing – original draft, Methodology. **Yang Wang:** Investigation, Formal analysis. **Ke Dong:** Investigation, Formal analysis. **Shilong Zhao:** Visualization, Formal analysis. **Miao Yang:** Writing – review & editing. **George Q. Huang:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the China Postdoctoral Science Foundation (2023M730406), the Natural Science Foundation Project of Chongqing (CSTB2024NSCQ-MSX0561), the National Natural Science Foundation (92367107), the Program of State Key Laboratory of Mechanical Transmission in Chongqing University (SKLMT-ZZKT-2024Z06).

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